Fuzzy Logic: Theory and Applications

Topics

- Introduction
- Conventional and fuzzy sets
- Basic concepts of fuzzy sets
- Fuzzy decision process
- Fuzzification
- Rule evaluation
- Defuzzification
- Fuzzy logic applications
- Summary
- Exercise

Note: Unit 8 is based on Motorola Fuzzy Logic Educational Program and Chapter 18 of "Decision Support Systems and Intelligent Systems", E. Turban and J. Aronson, Prentice Hall

Introduction

- The concept of Fuzzy Logic (FL) was conceived by Prof. Lofty Zadeh in the mid 1960s
- People do not require precise, numerical information input and yet they are capable of efficient decision making
- FL is a problem-solving methodology that simulates the process of normal human reasoning under *uncertainty*
- FL provides a simple way to arrive at a definite conclusion based upon vague, ambiguous, imprecise, noisy, or missing input information
- It can be implemented in hardware, software, or a combination of both
- Uses the mathematical *theory of fuzzy sets*

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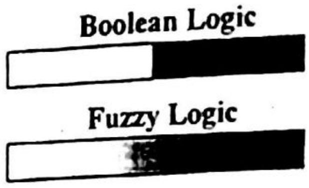
Conventional and Fuzzy Sets



- Boolean logic: true-false nature
- Infinite gradations between them
- *Fuzzy set* is a class of objects with a continuum of membership grades
- A *membership function*, which assigns to each object a grade of membership, is associated with each fuzzy set

Let U be a set of objects (elements) denoted by x, U = {x}

A fuzzy set A in U is characterised by a set of ordered pairs A = {(x), m_A (x)}, $x \in U$

where m_A is the membership function of x in A.

Membership function m_A(x) assumes its values in [0,1].

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- In **conventional set theory**, a set S in U is defined by a characteristic function $f_S(x)$ that maps elements of U to 1 *or* 0

$$f_S : S \rightarrow \{0,1\}$$

For an element x of U, $f_S(x) = \begin{cases} 1, & \text{if } x \in S \\ 0, & \text{if } x \notin S \end{cases}$

In **fuzzy set theory,** a fuzzy set A in U is defined by a a membership function $m_A(x)$, that maps elements of A to a value *between* 0 and 1

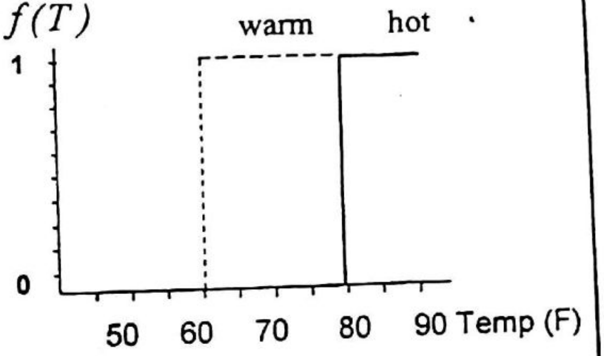
$$m_A : A \rightarrow [0,1]$$

For an element x of A, $m_A(x)$ is the degree to which x is an element of A

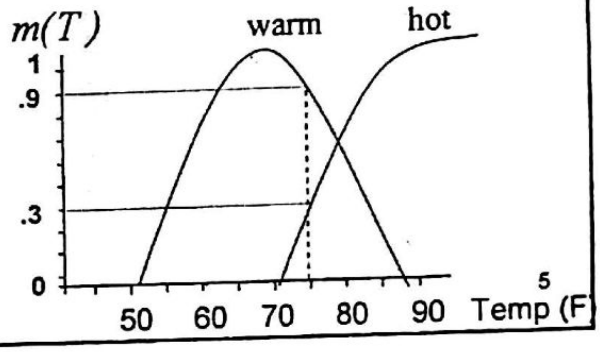
$m_A(x)=1$	x is *totally* in A
$0<m_A(x)<1$	x is *partially* in A
$m_A(x)=0$	x is *not* in A

Example:

- Is 80 degrees Fahrenheit warm or hot?
- In the real world "some of both" might be the answer



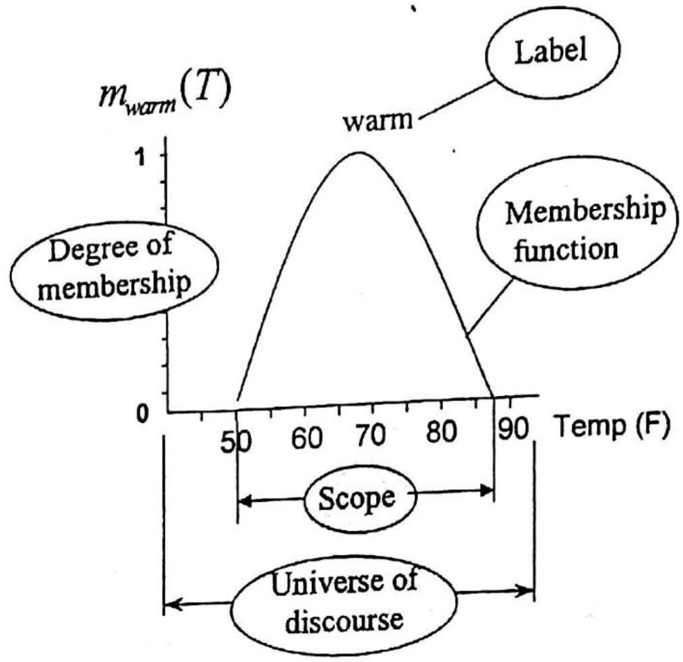
- *Conventional (Boolean) set*: Transition from set to set is *instantaneous* (an object is a member of set or it is not)



- With *fuzzy logic*, however, the transition from set to set is *gradual* (an object can have partial membership in multiple sets)

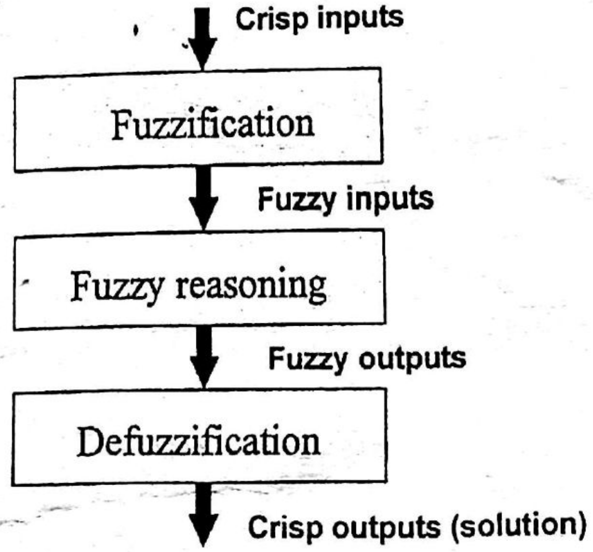
Basic Concepts of Fuzzy Sets

- Label: Descriptive name used to identify a membership function
- Degree of membership: the degree to which a crisp value (element) is compatible with a membership function
- Membership function: defines a fuzzy set by mapping crisp values from its domain to degrees to membership
- Scope (domain): width of the membership function (fuzzy set)
- Universe of discourse: range of all possible values applicable to a system variable



Fuzzy Decision Process

- Using fuzzy logic to reach *a crisp* solution to a specific decision or control problem usually involves three steps:

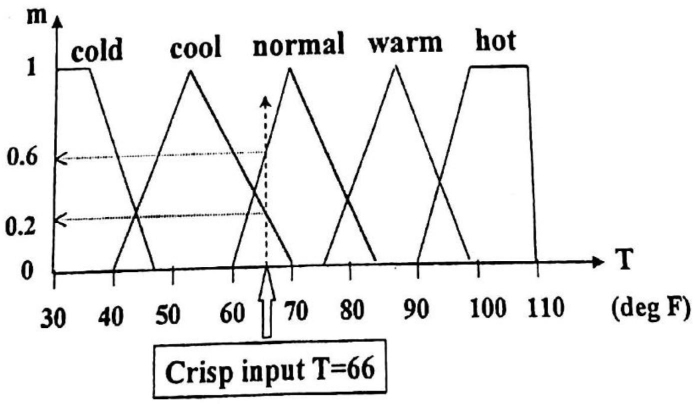


Fuzzification

- Assigns fuzzy labels in the Universe of Discourse to each of the *crisp inputs*
  - a crisp input can have multiple labels
  - the greater the number of labels, the higher the resolution; however it increases the computation time
- Next, membership functions are established to each label. It includes specifying
  - the domain
  - the shape of functions: trapezoidal, triangular, other
  - the degree of membership for each input value from the domain
- Finally, combining the crisp input with membership function to produce *fuzzy inputs*

Example

Crisp input (variable):	Outdoor Air Temperature T (deg F)				
Labels:	Cold	Cool	Normal	Warm	Hot
Domains:	30-47	40-70	60-84	75-98	90-110

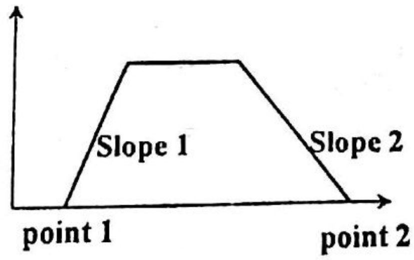


- For the crisp input T=66, there are two fuzzy inputs: T_cool, m=0.2 and T_normal, m=0.6

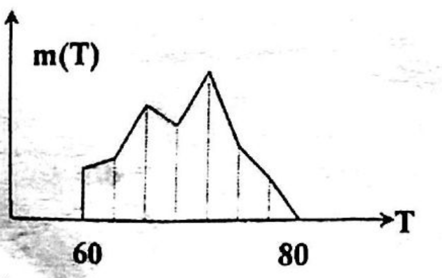
Membership Functions

- Membership functions can take several *different* shapes
- The shape affects the fuzzy processes in subtle ways
- *Trapezoidal and triangular* are most frequently used
- Other shapes require more complicated equations or large look-up tables
- *Singletons* (fuzzy sets that contain only one element) are easily represented in a computer and allow for simpler defuzzification algorithms
- Depending on the shape, various methods are used to represent a function in a computer

- *Point-slope representation:* allows trapezoidal, triangular and singleton membership function to be represented with a minimum amount of space and time



- A *look- up table* is a common representation for an arbitrary shaped function



Look-up Table

T	m(T)
60	0.25
61	0.29
62	0.43
....	

Fuzzy Reasoning (Rule Evaluation)

- Technique called *min-max inference* is generally used to calculate a numeric value representing the truth for a certain action or conclusion
- Also referred as *Rule Evaluation*, if fuzzy rules are applied
- Rules follow the common sense behaviour and are written in terms of the membership function linguistic labels
- Fuzzy rules are usually IF-THEN statements that describe action to be taken in response to various fuzzy inputs

  IF <antecedent 1> AND <antecedent 2> ...THEN <consequent >

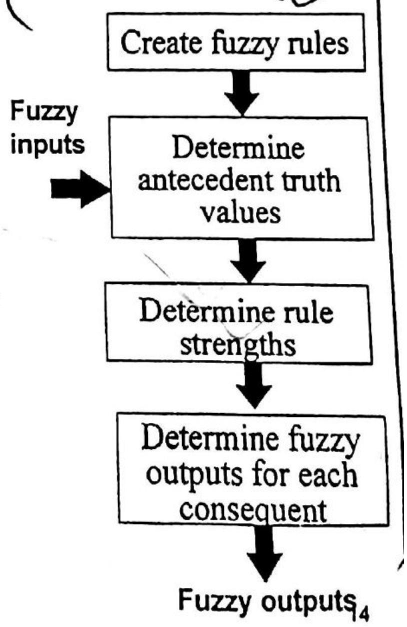
  <antecedent> : =< input_variable> is <label>

  <consequent> : =< output_variable> is <label>

- **Example:** IF the temperature is hot AND the soil is dry THEN the watering duration is long

Rule Evaluation Process

- Create *fuzzy rules* that describe the system behaviour
- For particular crisp inputs, determine the *degree of truth* of each antecedent using fuzzification
- Find the *strength of each rule* using fuzzy operation over its antecedents *(minimum)*
- Derive *fuzzy outputs* for all rule consequents, combining the strengths of the rules *(maximum)*
- The Rule Evaluation method used is *min-max inference*



Create Fuzzy Rules

- For a two-input, one-output system, the rules can be represented by the matrix

Moisture \ Temperature	Cold	Cool	Normal	Warm	Hot
Wet	short	short	short	short	short
Moist	short	med	med	med	med
Dry	long	long	long	long	long

- Sample rules extracted from the table:

IF temperature is cold AND soil is wet THEN watering duration is short

IF temperature is hot AND soil is dry THEN watering duration is long

- Such intuitive approach for system definition can replace a significant amount of mathematics, describing the underlying physics of the system

Determine Rule Strength

- Determine the *degree of truth* of each rule's antecedent (or degree of membership), corresponding to given crisp inputs
  - Extend a vertical reference line through each crisp input (x-value) and find the y-value where the line intersects the membership function
- Find the *rule strength* of each rule (rule's degree of truth)
  - for antecedents connected by AND operators, the rule strength assumes the *minimum* degree of membership (truth value) of the antecedents
  - for antecedents connected by OR operators, the rule strength assumes the *maximum* degree of membership (truth value) of the antecedents

Examples

IF x (truth value 0.3) AND y (truth value 0.7) THEN z **Rule Strength = 0.3** IF x (truth value 0.3) OR y (truth value 0.7) THEN z **Rule Strength = 0.7**

- However, it is recommended only AND operators to be used
- To convert the rules with OR operators

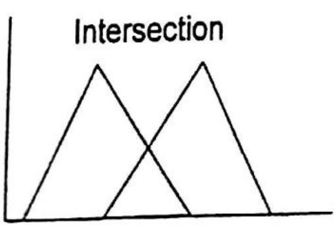
IF (x AND y) OR (v AND w) THEN z = IF x AND y THEN z IF v AND w THEN z

IF (x OR y) AND (v OR w) THEN z = IF x AND v THEN z IF x AND w THEN z IF y AND v THEN z IF y AND w THEN z

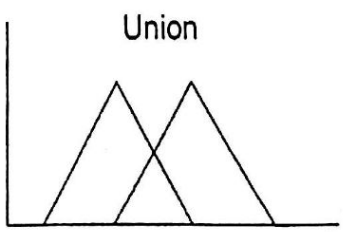
Fuzzy Operators

- The fuzzy **AND** operator corresponds to the *intersection* of fuzzy sets
- Intersection of two fuzzy sets is determined by taking the *minimum* of their membership functions over all x
- The fuzzy **OR** operator corresponds to the *union* of fuzzy sets
- The union of two fuzzy sets is determined by taking the *maximum* of their membership functions over all x

$$m_{A\cap B}(x) = \min[m_A(x), m_B(x)] \qquad m_{A\cup B}(x) = \max[m_A(x), m_B(x)]$$



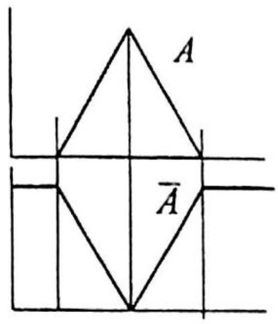
$A \cap B$



$A \cup B$

- The fuzzy **NOT** operator corresponds to the *complement* of a single fuzzy sets
- The complement $\overline{A}$ of a fuzzy set A is

$$m_{\overline{A}}(x) = 1 - m_A(x)$$



Example:

IF temperature is NOT hot AND soil is NOT dry
THEN watering duration is NOT long

Determine Fuzzy Outputs

- Fuzzy Output is calculated by comparing the strengths of all rules that specify *the same* consequent (action, conclusion)
- If multiple rules apply for an output action, the rule that is most true will dominate
- The Fuzzy Output is determined by a *maximum* operation

	Rule strength
R1: IF temperature is hot (.46) AND soil is dry (.25) THEN watering duration is long	.25
R2: IF temperature is warm (.2) AND soil is moist (.75) THEN watering duration is medium	.2
R3: IF temperature is warm (.2) AND soil is dry (.25) THEN watering duration is long	.2
R4: IF temperature is hot (.46) AND soil is moist (.75) THEN watering duration is medium	.46

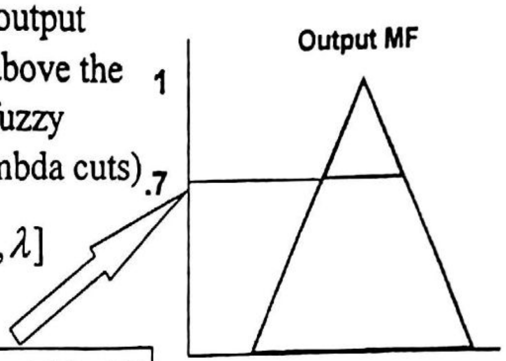
Fuzzy Output: Watering duration is .25 long and .46 medium

Defuzzification

- A process of combining all fuzzy outputs in a specific, crisp output of the system
- One of the mostly used defuzzification techniques is the Center of Gravity (COG) or centroid method

In COG method, each output membership function above the value indicated by its fuzzy output is truncated (lambda cuts)

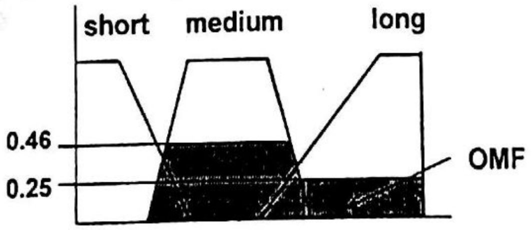
$$m_A(x) = \min[m_A(x), \lambda]$$



Fuzzy output (lambda) =0.7

- The resulting "clipped" membership functions are then combined in a complex Output Membership Function (OMF)

Watering time membership functions:	short	medium	long
Fuzzy output:	0	0.46	0.25



Watering time in minutes (Output)

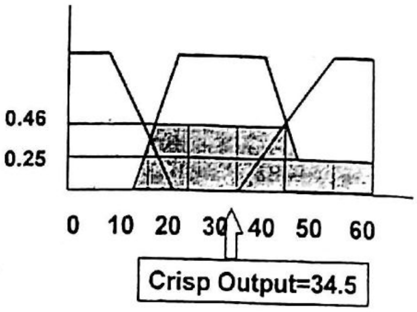
- The next step is to find the center of gravity COG of the OMF, which represents the *defuzzified (crisp) output*

□ COG formula

$$COG = \frac{\int_a^b m(x)x\,dx}{\int_a^b m(x)\,dx}$$

□ An estimate of the COG over a sample of points with a small step size

$$COG = \frac{\sum_{x=a}^{b} x.m(x)}{\sum_{x=a}^{b} m(x)}$$



$$COG = \frac{15(.3)+25(.46)+35(.46)+45(.46)+55(.25)}{.3+.46+.46+.46+.25} = \frac{66.55}{1.93} = 34.5$$

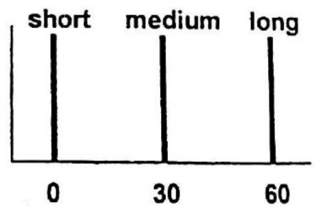
Singleton Defuzzification

- □ A singleton is represented by an individual point in the output space
- □ Truncation of singletons results in reduction of their heights
- □ COG defuzzification method can also apply to singleton output membership functions
- □ Calculations are significantly less
- □ The COG formula is

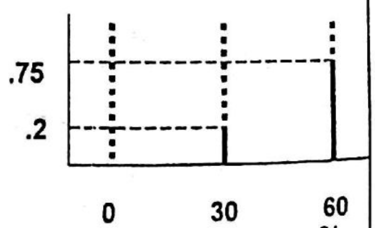
$$COG = \frac{\sum_i FO_i \times SPos_i}{\sum_i FO_i}$$

FO - Fuzzy Output

SPos - Singleton Position on x axis



	short	medium	long
FO:	0	0.2	0.75



Fuzzy Logic Advantages

- Provides flexibility
- Models uncertainty
- Allows for observation
- Shortens system development time
- Increases the system's maintainability
- Uses less expensive hardware
- Handles control or decision-making problems not easily defined by mathematical models

Fuzzy Logic Applications

- Strategic planning (Hall, 1987)
- Real estate appraisal
- Bond evaluation (Chorafas, 1994)
- Selecting stocks (on Japanese Nikkei Stock Exchange)
- Regulating auto antilock braking systems
- Automating laundry machine operation
- Building environmental controls
- Controlling video camcorders image position
- Controlling train motion
- Inspecting beverage cans for printing defects
- Keeping space shuttle vehicles in steady orbit
- Regulating shower head water temperature
- Controlling cement kiln oxygen levels
- Increasing industrial quality control application accuracy and speed
- Decision making (see Glenn [1994])

Summary

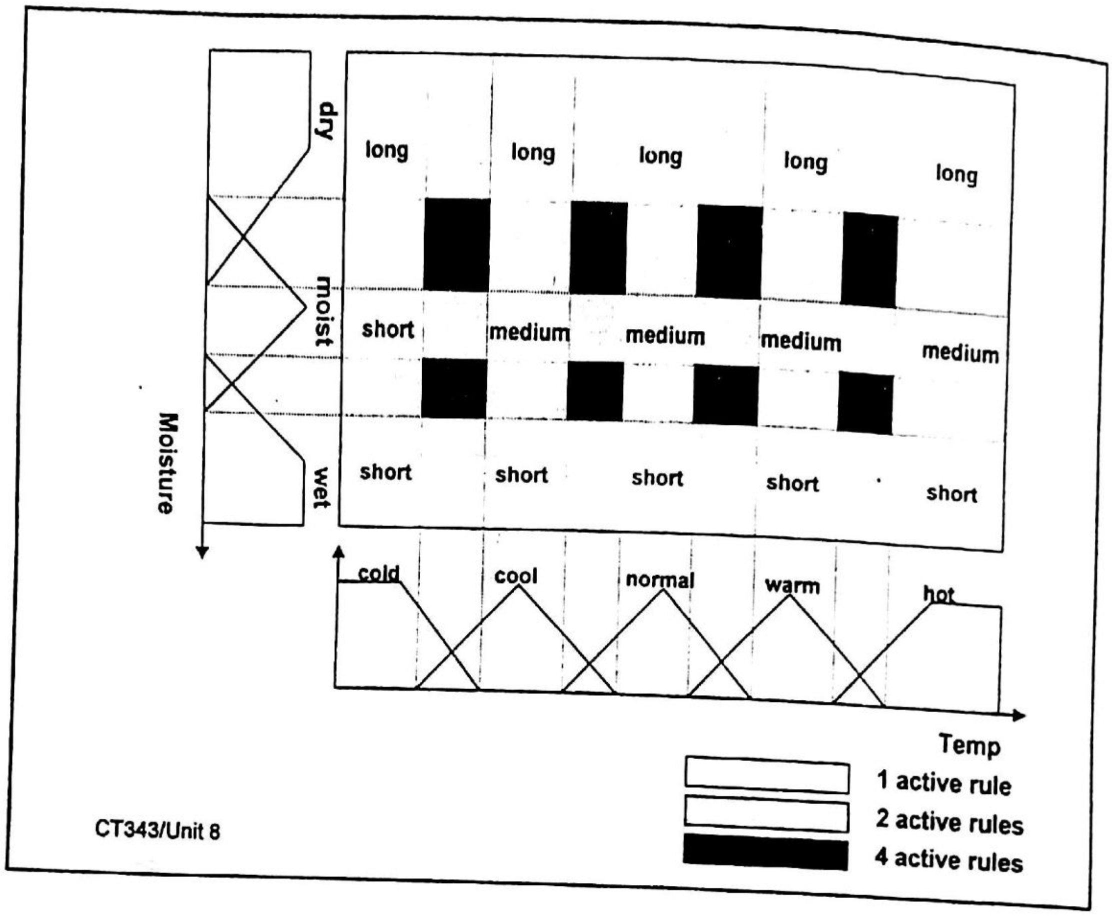
- Fuzzy Logic is a problem-solving methodology that simulates the process of normal human reasoning under *uncertainty*
- Fuzzy Logic uses the mathematical *theory of fuzzy sets*
- Fuzzy set is a class of objects with a continuum of membership grades
- A membership function, which assigns to each object a grade of membership, is associated with each fuzzy set
- Other elements of fuzzy sets are: Label, Degree of membership, Scope (domain) and Universe of discourse
- Fuzzy decision making uses fuzzy logic to reach a crisp solution to a specific decision or control problem

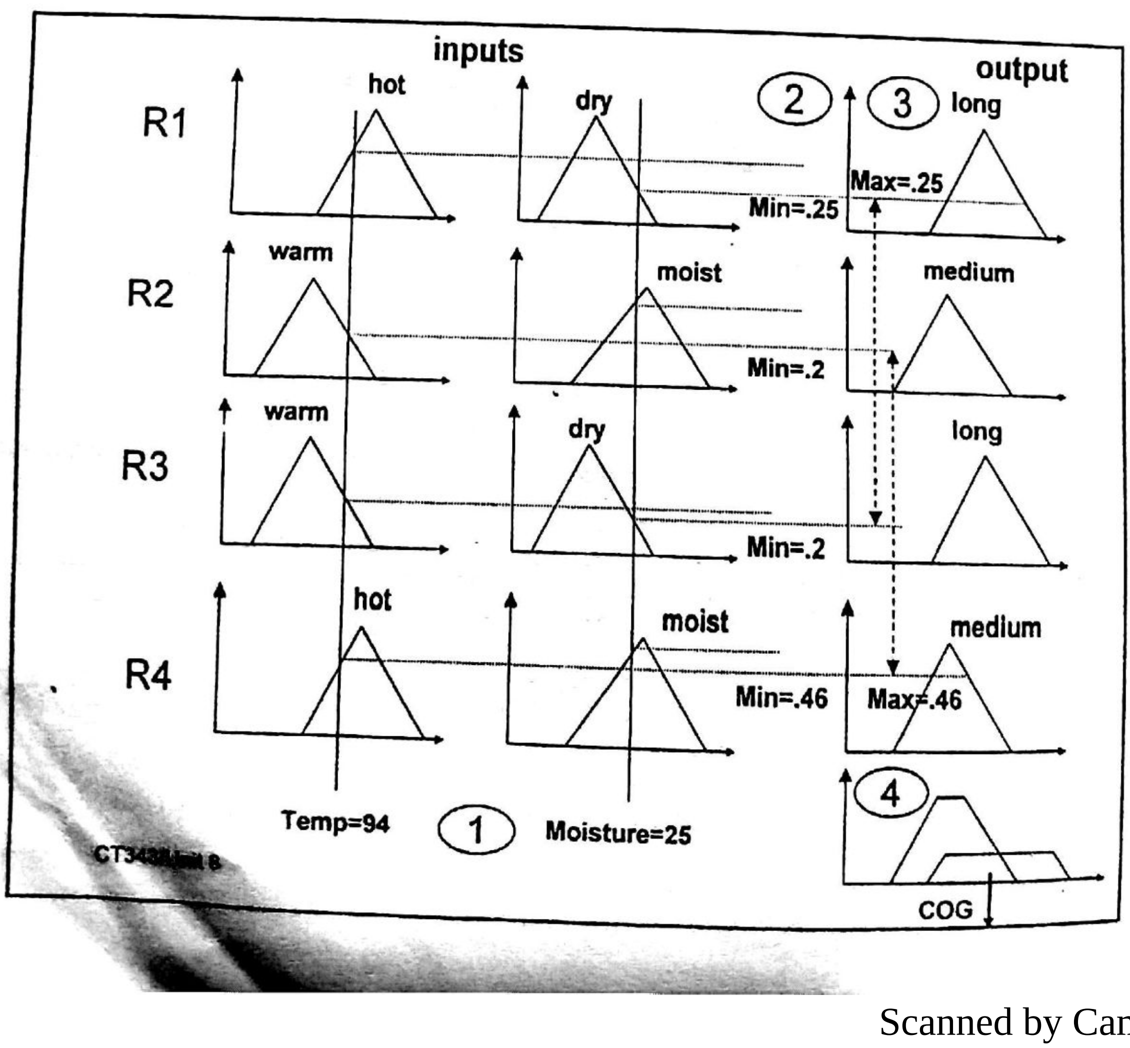
- Fuzzy decision making usually involves three steps
  - Fuzzification
  - Fuzzy reasoning
  - Defuzzification
- Fuzzification is a process of combining the crisp inputs with membership functions to produce fuzzy inputs
- Membership functions can take several different shapes that affect the fuzzy processes in subtle ways
- Depending on the shape, various methods are used to represent a membership function in a computer
- Technique called min-max inference is generally used to calculate a numeric value representing the truth for a certain action or conclusion
- Fuzzy rules are usually IF-THEN statements that describe action to be taken in response to various fuzzy inputs

- Fuzzy operators are used to determine the degree of truth (rule strength) for each rule, corresponding to given crisp inputs
- Fuzzy Output is calculated by comparing the strengths of all rules that specify the same consequent (action, conclusion), by a maximum operation
- Defuzzification is a process of combining all fuzzy outputs in a specific, crisp output of the system
- One of the mostly used defuzzification techniques is the Center of Gravity (COG) or centroid method
- COG defuzzification method can also apply to singleton output functions, which leads to significantly less calculations
- Fuzzy logic has some advantages
- Fuzzy logic is implemented in many areas of decision making, control, pattern recognition

Exercise

- Run the Fuzzy Logic Educational Program (PC Lab menu) and study the basics of Fuzzy Logic theory
- Study the Fuzzy Knowledge Builder tool from the PC Lab menu
- Using this tool, develop a fuzzy sprinkler control system based on the examples of this lecture and the information given in the Educational Program
- Provide a short report, describing your fuzzy system, its inputs and output, the membership functions, the rules and its performance


dry
moist
wet
Moisture
long
long
long
long
long
short
medium
medium
medium
medium
short
short
short
short
short
cold
cool
normal
warm
hot
Temp
1 active rule
2 active rules
4 active rules
CT343/Unit 8


inputs
output
R1
hot
dry
2
3
long
Max=.25
Min=.25
R2
warm
moist
medium
Min=.2
R3
warm
dry
long
Min=.2
R4
hot
moist
medium
Min=.46
Max=.46
4
Temp=94
1
Moisture=25
COG

Genetic Algorithms, Soft Computing and Neuro-Fuzzy Systems

Topics

- Genetic Algorithms
- Primary operations of the GA
- Soft Computing
- GA for Fuzzy Systems
- Co-operative Neuro-Fuzzy Systems
- Hybrid Neuro-Fuzzy Systems
- Applications of Neuro-Fuzzy Systems
- Summary

Note: Unit 9 is based on Chapter 18 of "Decision Support Systems and Intelligent Systems", E. Turban and J. Aronson, Prentice Hall and the course on Neural Fuzzy Systems, http://www.abo.fi/~rfuller/nfs.html

Genetic Algorithms

- *Genetic algorithms* (GA) are search algorithms based on the mechanics of natural selection and natural genetics
- GA are also known as evolutionary algorithms
- Basic Goal is to demonstrate *Self-organisation* and A*daptation* by exposure to the environment
- GA System *learns* to adapt to changes
- GA have been developed by John Holland and his colleagues at the University of Michigan in the beginning of 1970s
- The goal of their research was twofold:
  - to explain the adaptive processes of natural systems
  - to design AI systems software that retains the important adaptation mechanisms of natural systems
- This research has led to important discoveries in AI science

Definitions and Process

- *Genetic algorithm*: "an iterative procedure maintaining a population of structures that are candidate solutions to specific domain challenges (Grefenstette [1982])
- The smallest unit of a GA is called a *gene*. It represents a unit of information in the problem domain
- A series of these genes, or a *chromosome* represents one candidate solution

Examples:

For a **routing problem** a chromosome might look like

New York	London	Paris	......	Berlin

Interpretation: Start in New York, proceed to London, then to Paris, ...etc.

For a **portfolio problem,** a chromosome might look like

3	5	7	2	......	4

Interpretation: Buy 3% of Stock A, 5% of Stock B, ...etc.

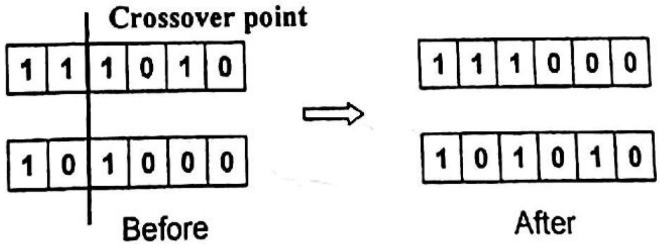
Primary Operators of the Genetic Algorithms

"You can't always get what you want....
but if you try sometimes, you just might find...
you get what you need." Mick Jagger

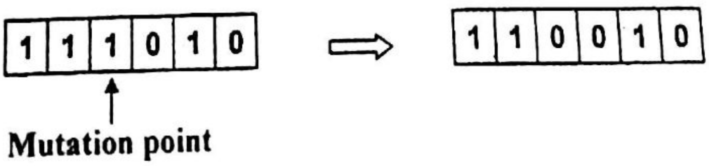
- Chromosomes are represented as *strings of binary symbols*
- GA creates an initial *population* of chromosomes
- Iterative process of *refining the* initial solutions
- A *fitness function* determines which chromosome solutions are good and which are bad
- GA uses three genetic operators: *reproduction (selection), crossover and mutation*
- These operators produce new chromosomes which form a new *population (generation)*

- *Reproduction* is a process in which the chromosomes are selected according to their *fitness function* value
- Through reproduction, GA produces new generations of improved solutions by selecting parents with higher fitness ratings
- This operator is a version of natural selection, a Darwinian survival of the fittest among all population
- *Crossover* involves the exchange of information between two chromosomes, selected as parents
- During crossover, the chromosomes swap some of their genes
- Crossover may proceed in two steps:
  - selected chromosomes are mated at random
  - each pair of chromosomes undergoes crossing over, choosing a random position of the string and exchanging the segments to the right and to the left of this position

- Two new strings are created by this swapping, which are members of the new generation



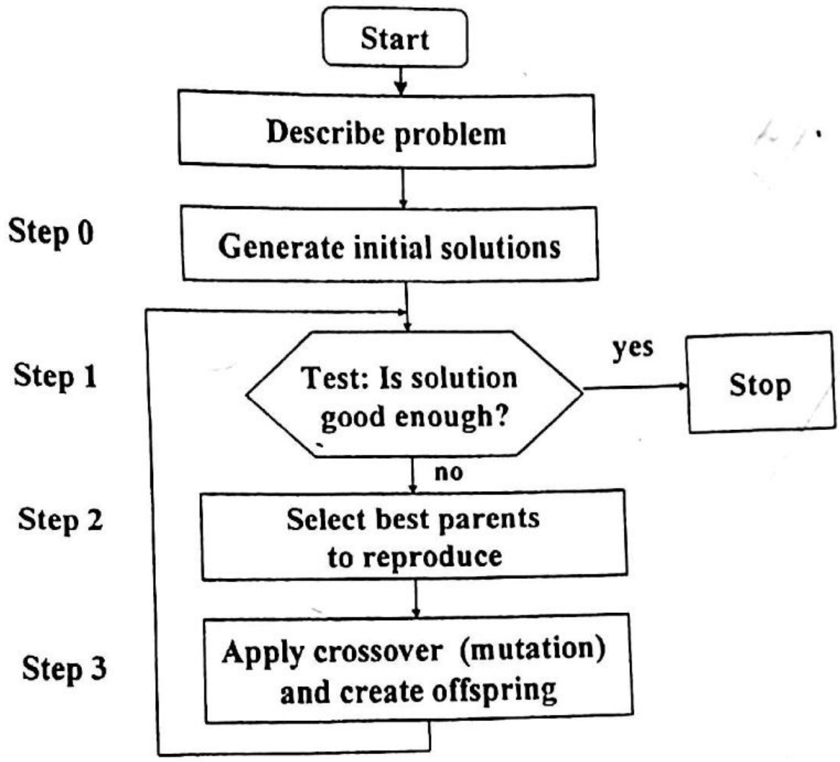
- *Mutation* is an arbitrary change of the value of a string, it simply changes 1 to 0 or 0 to 1 at a particular location of the string
- Mutation plays secondary role in the GA operations. Such a change occurs with a very small rate



Example: Vector Game

- You play this game against an opponent who secretly writes down a string of six digits, each digit is 0 or 1
- You must try to guess this number as quickly as possible
- You present a number to your opponent and he/she will tell you how many of your digits are correct (but not which ones)
- Solutions:
  - Random Trial and Error: 64 possible strings
  - Genetic Algorithm Solution (see next Table)

Secret Number	0	0	1	0	1	0	
Initial solutions							score
A	1	1	0	1	0	0	1
B	1	1	1	1	0	1	1
C	0	1	1	0	1	1	4
D	1	0	1	1	0	0	3
Select C and D as parents.	Crossover C and D after the second digit:						
C	0	1	1	0	1	1	4
D	1	0	1	1	0	0	3
E (C[1,2], D[3,..,6])	0	1	1	1	0	0	3
F (D[1,2], C[3,..,6])	1	0	1	0	1	1	4
	Crossover C and D after the fourth digit:						
G (C[1,..,4], D[5,6])	0	1	1	0	0	0	4
H (D[1,..,4], C[5,6])	1	0	1	1	1	1	3
Select F and G as parents.	Crossover F and G after the first digit:						
I (F[1], G[2,..,6])	1	1	1	0	0	0	3
J (G[1], F[2,..,6])	0	0	1	0	1	1	5
	Crossover F and G after the third digit:						
K (F[1,2,3], G[4,5,6])	1	0	1	0	0	0	4
L (G[1,2,3], F[4,5,6])	0	1	1	0	1	1	4
Select J and K as parents.	Apply a crossover after the fifth digit						
M (J[1,..,5], K[6])	0	0	1	0	1	0	6
N (K[1,..,5], J[6])	1	0	1	0	0	1	3



- Fig 18.11. Flow Diagram of the Genetic Algorithm Process

General Areas of Genetic Algorithm Applications

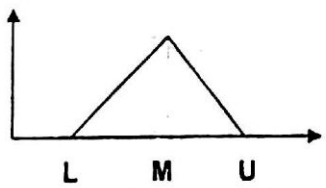
- Dynamic process control
- Induction of rule optimisation
- Discovering new connectivity topologies
- Simulating biological models of behaviour and evolution
- Complex design of engineering structures
- Pattern recognition
- Scheduling
- Transportation
- Layout and circuit design
- Telecommunication
- Graphs

Soft Computing

- Neural Networks, Fuzzy Systems and Genetic Algorithms are categorised into a same computational paradigm, called Soft Computing
- Each technology has different advantages and disadvantages
- Neural networks are good at recognising patterns, but not good at explaining how they reach their decisions
- Fuzzy logic systems can reason with imprecise information, explain their decisions but they cannot automatically acquire the rules they use to make those decisions
- GA provide powerful methods for optimising complex problems but require a lot of computations
- Intelligent integrated (hybrid) systems: where two or more techniques are combined in a manner that overcomes the limitations of individual techniques
- Soft Computing tries to combine these technologies

Genetic Algorithms for Fuzzy Systems

- GA is capable of designing the membership functions and their parameters
- Two main issues to be resolved: a genetic coding of parameters and a method for determining the fitness function



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Membership function chromosome (MFC)

- Example: IF x1 is small and x2 is big THEN y is medium

x1=small: MFC1
x1=big: MFC2
x2=small: MFC3
x2=big: MFC4
y=small: MFC5
y=medium: MFC6

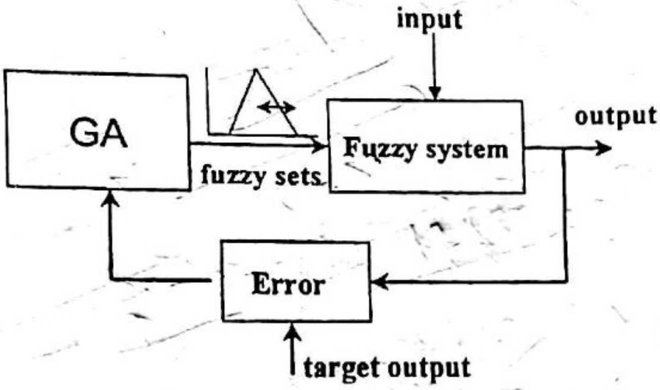
MFC1	MFC4	MFC 6

Composite rule chromosome

- Determining the fitness function

$$S = \sum_{i=1}^{N} (target_i - output_i)^2$$

N - number of input-output data pairs



- Similar approach can be used for determining the weights of NN (GA-based weight learning)

Properties of fuzzy systems and neural networks

Skills		Fuzzy Systems	Neural Nets
Knowledge acquisition	Inputs	Human experts	Sample sets
	Tools	Interaction	Algorithms
Uncertainty	Information	Quantitative and Qualitative	Quantitative
	Cognition	Decision making	Perception
Reasoning	Mechanism	Approximate	Parallel computations
	Speed	Low	High
Adaptation	Fault-tolerance	Low	Very high
	Learning	No	Adjusting weights
Natural language	Implementation	Explicit	Implicit
	Flexibility	High	Low

19 Neuro Fuzzy Systems

Neuro-Fuzzy Systems

- To incorporate the concept of fuzzy logic into the neural networks. The resulting integrated system is called neuro-fuzzy system or fuzzy-neuro network
- Two types of combinations between NN and FS
  - Co-operative neuro-fuzzy systems: both work independently of each other
  - Hybrid neuro-fuzzy systems: both are combined into a homogeneous architecture
- Co-operative approaches intend to specify parameters of an ordinary fuzzy system using learning methods obtained from neural network
- Hybrid approaches intend to apply a learning algorithm to a fuzzy system by representing it in a special neural-network-like architecture

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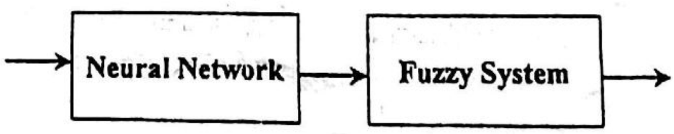
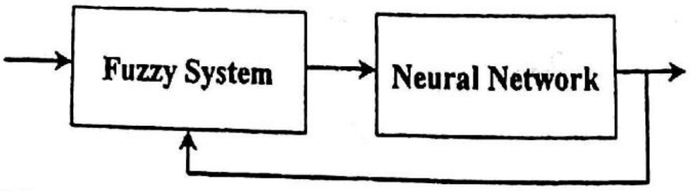
Cooperative Neuro-Fuzzy Systems

- NN are used to process inputs or outputs of FS:
  - to preprocess input data (off-line): fuzzy rules extraction and learning the membership functions



  - to postprocess the fuzzy system outputs and to tune the membership functions (on-line)



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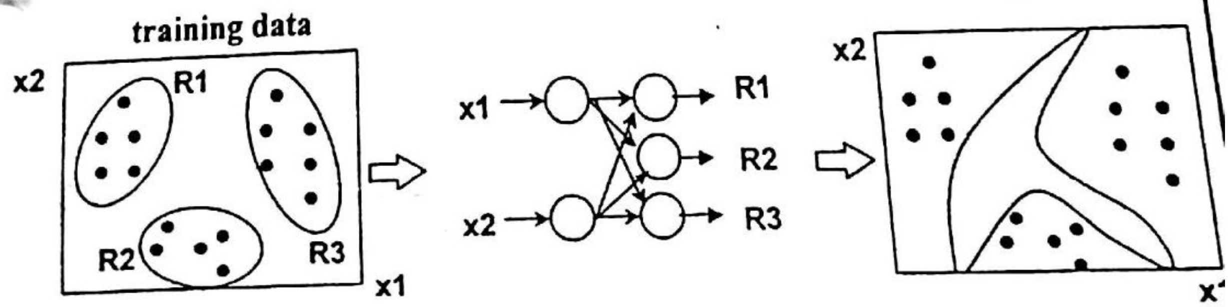
Fuzzy Rules Extraction (preprocessing input data)

- For neural networks, the knowledge is *automatically* acquired, but it is not possible to extract structural knowledge (rules) from the trained NN (*black box*)
- For fuzzy systems, knowledge is represented as fuzzy rules, but knowledge acquisition is difficult
- NN can automatically extract fuzzy rules from numerical data, using a clustering approach
- It usually takes a lot of time to design and tune the membership functions of fuzzy systems
- Neural network learning techniques can automate this process
- NN learns the membership functions from training data by determining suitable parameters

- NN uses the training data to specify the fuzzy rules and the membership degrees for given input values
- The main steps are:
  - Clustering the given training data and deciding the number of rules
  - Training and partitioning the input space by NN
- After training, the NN computes the degrees to which an input vector belongs to each cluster (its membership values)



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Fuzzy Partitioning

- **Example:** Sprinkler fuzzy control system
- The membership functions of temperature and soil moisture are designed independently
- The resulting fuzzy partitioning of the input space resembles Fig 1
- However, when the input variables are dependent, fuzzy partitioning such as Fig 2 is more appropriate

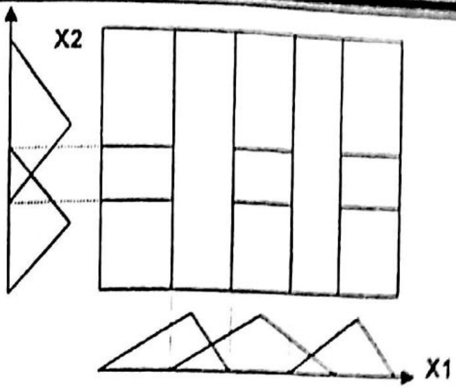

Fig 1. Fuzzy partitioning

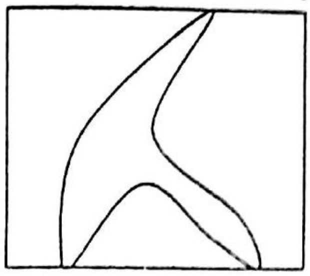

Fig 2. Desired partitioning

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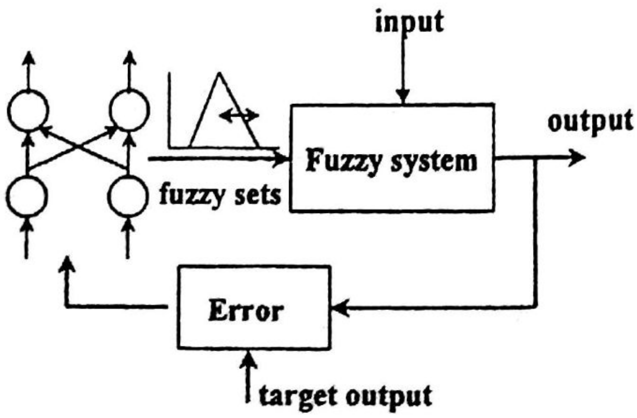
Tuning Membership Functions (postprocessing)

- NN learns parameters on-line (during the use of the FS) to adapt membership functions
- Fuzzy rules and initial membership functions are given
- The learning process is guided by the error between the FS output and target output



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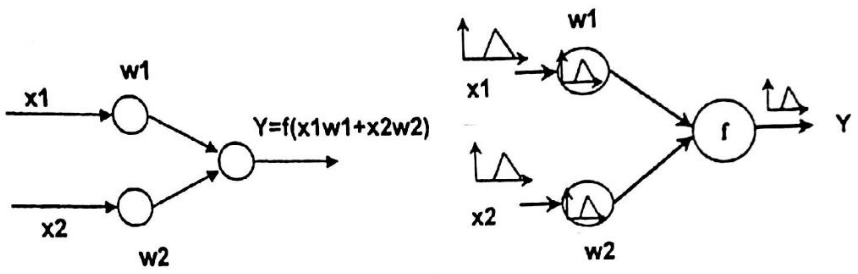
Hybrid Neuro-Fuzzy Systems

- The main idea is to interpret the rule base of a FS in terms of a NN (fuzzification of conventional NN)
- A processing element of a hybrid system is called a *fuzzy neuron* (similar to the artificial neuron)
- All signals and weights are fuzzy sets (fuzzy numbers)



Artificial neuron **Fuzzy neuron**

- Hybrid NN can be used to implement fuzzy IF-THEN rules
- The fuzzy rules are represented by combination of AND and OR fuzzy neurons
- AND fuzzy neuron: The inputs and weights are combined by max (OR) operation, the signals V are aggregated by min (AND) operation

$$V_i = max(w_i, x_i), i = 1,2,...,n$$

$$Y = min(V_1...V_n) = min(max(x_1, w_1)...max(x_n, w_n))$$

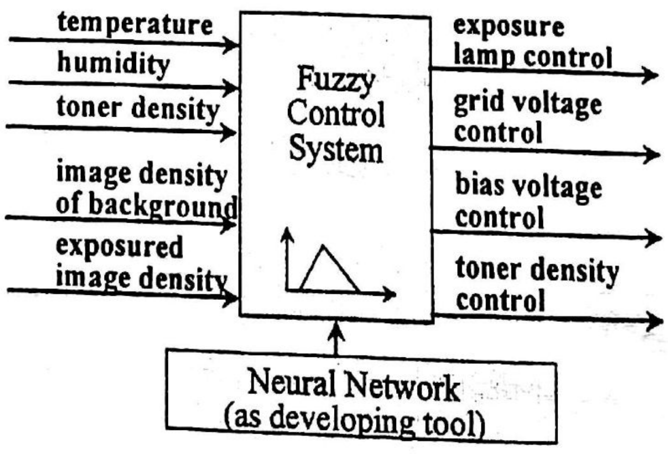
- The AND neurons realise the min-max composition, used in fuzzy rule evaluation
- OR fuzzy neurons realise the max-min composition
- The learning algorithm modifies the fuzzy weights in a supervised mode

Applications of Neuro-Fuzzy Systems

- The first applications of fuzzy neural networks to consumer products appeared on the Japanese market in 1991
- Some applications: air conditioners, electric fans, electric heaters, microwave ovens, refrigerators, vacuum cleaners, washing machines, photocopying machines, word processors
- Application areas:
  - control
  - classification
  - prediction
  - pattern recognition

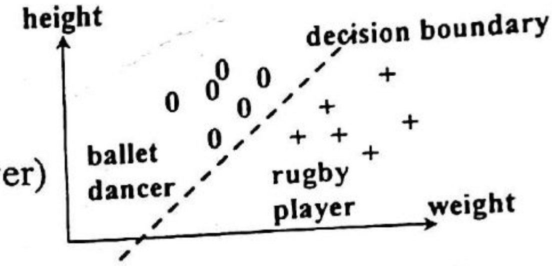
Neuro-Fuzzy Control

- Example: Photocopier machine (Matsushita Electric)



Neuro-Fuzzy Classification

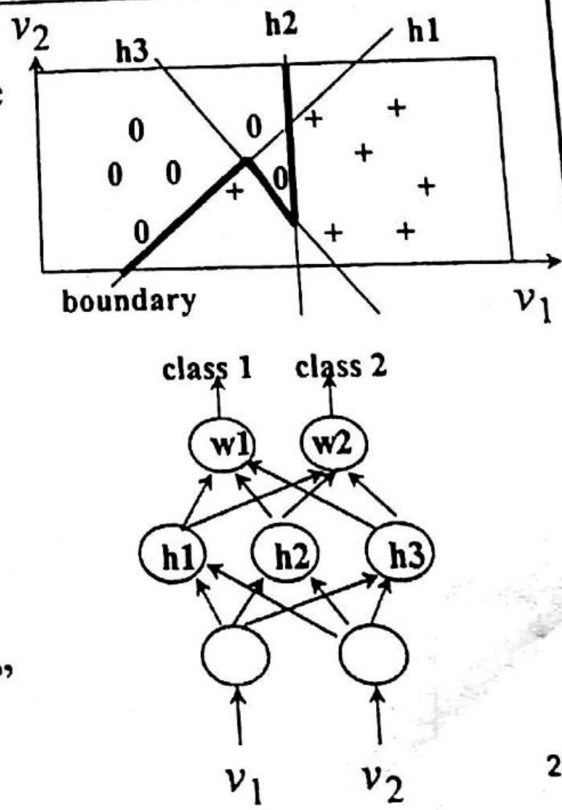
- Pattern Classification is an intelligent behaviour
- *Pattern=[v,w]; v - observations; w - category (class)*
- Pattern Classification is a process of inferring *w* from *v*
- Classification tries to find decision boundaries between classes
  - Example:
    v=(height, weight)
    w=(ballet dancer, rugby player)



- Conventional approaches of classification involve clustering training samples and associating clusters to given categories
- These statistical methods lack an effective way of defining the boundaries among clusters

NN Classification

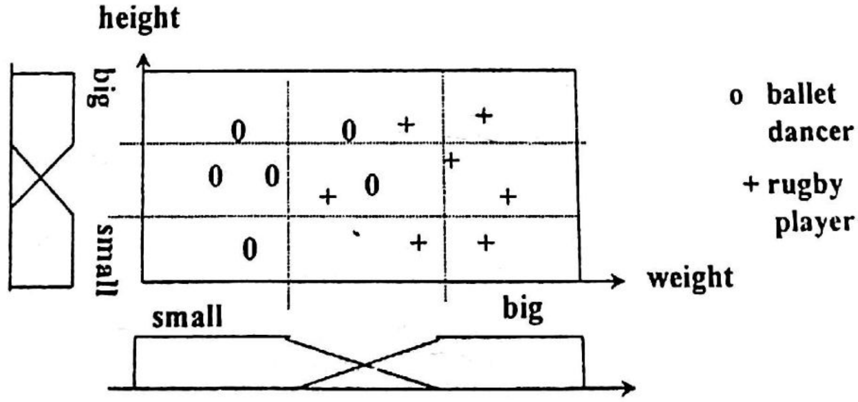
- NN can deal with non-linear classification problems because it can form more complex decision boundaries
- Each node in the hidden layer can create a linear boundary
- Each node in the output layer combine the lines to create convex decision region
- A NN can classify the input vector in different classes, by setting its outputs to "1" or "0"



Fuzzy Logic Classification

- Fuzzy classification assumes the boundary between two neighbouring classes as a continuous, overlapping area, within which an object has partial membership in each class
- The task of fuzzy classification is to generate an appropriate fuzzy partition of the feature space (appropriate means that the number of misclassified patterns is very small or zero)
- If fuzzy partition is not set correctly, or if the number of linguistic terms for the input features is not large enough, then some patterns would be missclassified
- If the system is very complex, great number of rules are needed
- No formal methods to tune the membership functions exist

- Classification rules

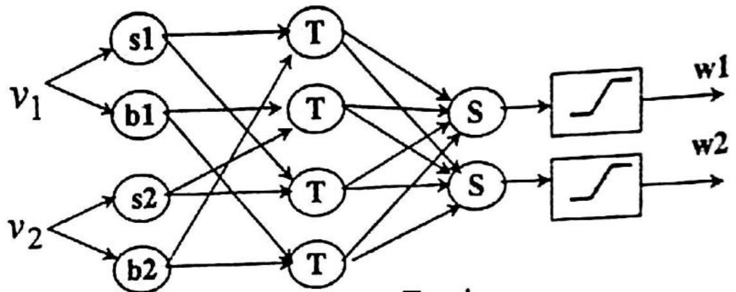


IF weight is big and height is big THEN the pattern is a player
IF weight is small and height is big THEN the pattern is a dancer
IF weight is big and height is small THEN the pattern is a player
IF weight is small and height is small THEN the pattern is a dancer

Neuro-Fuzzy Classifier

- Adaptive Network-based Fuzzy Inference System ANFIS
- Architecture with two input variables, each one with two linguistic labels; training data are categorised by two classes
- MIN-MAX inference, using 4 rules:

  IF v_1 is A_{1i} and v_2 is A_{2i} THEN pattern is w_i



T=min
S=max

s=small
b=big

Summary

- *Genetic algorithms* (GA) are search algorithms based on the mechanics of natural selection and natural genetics
- The smallest unit of a GA is called a *gene*
- A series of genes, or a *chromosome* represents one candidate solution
- A *fitness function* determines which chromosome solutions are good and which are bad
- GA uses three genetic operators: *reproduction (selection), crossover and mutation*
- NN, FS and GA categorised into a computational paradigm, called Soft Computing
- Soft Computing tries to combine the advantages of these technologies